SYSTEMATIC STRATEGIES · VALUE

Cheap markets, cheaper stocks

A value strategy built at two levels — ETFs on cheap countries and sectors, then the cheapest stocks inside them — beat its value benchmark over twenty years with little more than half of its worst loss. The stocks did the heavy lifting, and the clean test only starts now.

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Value investing has two problems. The first is timing: markets become cheap because they are falling, and in a crash they keep falling together. The second is precision: a sector can be cheap while many of the companies inside it are not.

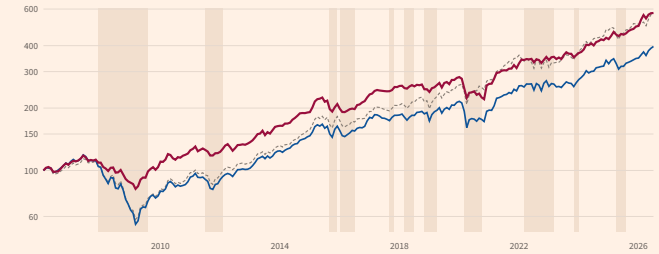
The Global Value Rotation, the official value strategy published on Thoth Ledger, answers both. It holds 70% in ETFs on the cheapest developed countries and sectors, 30% in the eight cheapest stocks inside them, and halves the whole portfolio when the market turns. From February 2006 to June 2026 it returned 8.9% a year net of costs, against 7.0% for its benchmark, MSCI World Value in euros, and 8.9% for MSCI World. Its worst drawdown was −31%, against −54% and −49%; its Sharpe ratio 0.72, against 0.43 and 0.58.

The lead holds in both halves of the period, but it is not statistically significant: the t-statistic of the monthly excess return is 1.09. The strategy beat its benchmark in 11 calendar years out of 21. It matches MSCI World on return over twenty years, not since 2016.

Those numbers deserve as much suspicion as interest. This note explains how the strategy works, where its return comes from, where it fails, and why the only test that really counts started this month.

Growth of 100 invested at the end of January 2006

Log scale, EUR, net of costs. Shaded: months at 50% exposure.



Global Value Rotation MSCI World Value Net EUR MSCI World Net EUR

Source: Bloomberg, Thoth Ledger

Two levels, one risk rule

The core (70%). At each month end, 25 indices are scored: 14 developed countries and the 11 sectors of MSCI World. The score averages three z-scores — earnings yield, book yield and dividend yield — each measured against the index's own ten-year history. The four cheapest countries and four cheapest sectors are held through one UCITS ETF each; a line already held stays until it leaves the six cheapest.

The satellite (30%). Among the 300 largest stocks of MSCI World at each date — including those later delisted or taken over — the satellite ranks each company within its sector on EV/EBIT, book-to-price, free-cash-flow yield and sales-to-EV. Accounts are used only six months after the year end. It holds the eight cheapest stocks in the countries or sectors held by the core, at most two per sector and three per country, and keeps a stock while it stays in the cheapest 40% of its sector.

The risk rule. Three indicators are checked at each month end: MSCI World below its ten-month average, euro high yield lagging short government bonds, fewer than half of the 25 lines in an uptrend. When at least two are red, the whole portfolio is cut to 50% equities, never to zero, until fewer than two are red.

The costs. The backtest charges 0.10% per unit of turnover and 0.25% a year of fees on ETFs, 0.15% per unit of turnover and 0.40% a year of withholding tax on stocks.

Student research by Noah Doose. Thoth Ledger is not an investment firm and offers no product or service. Hypothetical backtest of a model portfolio on monthly total returns in EUR (indices as ETF proxies, stocks with dividends), net of the costs stated, before taxes. Data: Bloomberg (MSCI indices and compositions, valuation ratios, stock returns and fundamentals, euro money-market rates). Past performance does not guarantee future returns. Not investment advice.

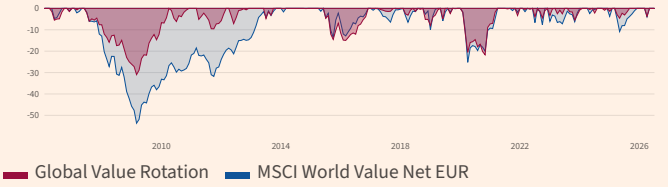
Where the return comes from

Mostly from the stocks. On its own, the satellite earned 9.8% a year (t = 1.34 against MSCI World Value); the ETF core earned 8.5% (t = 0.72). At country and sector level, the value score's twelve-month rank information coefficient was close to zero (−0.05 for countries, −0.03 for sectors). A sector ETF holds hundreds of companies and they cannot all be cheap: ranking stocks within their sector is where the signal is sharp enough to pay.

The core brings the risk control. The risk rule held the portfolio at 50% in 58 of 245 months, in 28 switches. In 2008 the strategy lost 20.0% while MSCI World Value lost 37.7%; in 2022 it gained +3.8% while MSCI World lost 12.8%.

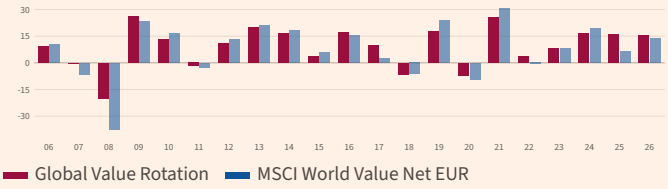
The rule has a cost, paid in quick rebounds: −7.2% in 2020 against +6.3% for MSCI World. And when expensive growth leads, value lags whatever the rule: +8.1% in 2023 and +16.6% in 2024, against +19.6% and +26.6%.

Drawdown from previous peak (%)



Calendar-year returns (%)

06-26 = 2006-2026; 2026 to June.



Two halves, two stories

The research was organised around a design half (2006-2015) and a validation half (2016-2026). In the first, the strategy earned 7.2% a year against 4.6% for MSCI World Value, mostly by losing less in 2008. In the second, 10.7% against 9.3%, with the satellite at 12.2%.

The benchmark is MSCI World Value because the strategy is a value strategy. MSCI World earned the same 8.9% a year over twenty years, but much more since 2016 (12.0% a year), so it stays on every page as the investor's real alternative. Against it the strategy offers about the same return with a worst loss of −31% instead of −49%.

THE NUMBERS

Holdings	**8 ETFs + 8 stocks**
Weights	**8.75% per ETF, 3.75% per stock**
Risk rule	**100% or 50% invested**
Rebalancing	**Monthly**
Benchmark	**MSCI World Value Net EUR**
Backtest	**Feb 2006 – Jun 2026, EUR**
CAGR	**8.9%**
Volatility	**10.9%**
Sharpe	**0.72**
Max drawdown	**−31%**
t vs benchmark	**1.09**
Months at 50%	**58 / 245**

NEXT TO HORUS SMART BETA

The value strategy can sit next to Horus Smart Beta 60/40, the multi-asset strategy of Horus AM. In euros the two are correlated at 0.67, and only 0.46 in months when world equities fall, because Horus keeps 40% in bonds. With the dollar taken out, both earn about the same, so a mix lowers risk rather than raising return: the Sharpe ratio rises to about 0.77, from 0.71 for value alone. Unhedged, Horus would show 9.9% instead of 8.6%: the gap is the dollar, not the strategy.

Mix	CAGR	Sharpe	Max DD
100% value / 0% Horus	8.9%	0.71	−31%
70% value / 30% Horus	8.9%	0.76	−25%
50% value / 50% Horus	8.8%	0.77	−23%
30% value / 70% Horus	8.8%	0.75	−22%
0% value / 100% Horus	8.6%	0.68	−20%

Horus hedged into EUR (strategy performance only), 2007-01 to 2026-06, rebalanced monthly.

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How it was built — and why that matters

The first rules were fixed on 30 October 2026, before any data were extracted: four countries and four sectors, no risk rule. They earned more than the benchmark with almost the same 50% drawdown. Each later step was written down and committed to the code repository before it was run: a holding buffer and a trend filter (V4c), a three-sector version (V5), the 50% risk rule instead of all-or-nothing cash (V7), and finally the stock satellite, with its universe, score and rules fixed before the first result.

Each step was chosen after seeing earlier results over the whole period, so 2016-2026 is not a clean out-of-sample test, and a reader should haircut the Sharpe ratio for the number of trials (Bailey and López de Prado, 2014). The clean test is the live record from November 2026; the monthly factsheet will report it without restating the backtest.

The main versions, Feb 2006 – Jun 2026

Version	CAGR	Sharpe	Max DD	2006-15	2016-26
V1 Rules of 30 Oct: 4 countries + 4 sectors	7.7%	0.49	−50%	4.7%	10.6%
V4c Buffer + 100% cash trend filter	8.5%	0.78	−19%	9.1%	7.8%
V5 3 sectors only, 50% de-risking	8.5%	0.60	−31%	7.4%	9.6%
V7 ETF core: V4c, cut to 50%	8.5%	0.70	−31%	7.0%	9.9%
Satellite: 8 cheapest stocks	9.8%	0.64	−32%	7.3%	12.2%
Strategy 70/30 (official)	**8.9%**	**0.72**	**−31%**	**7.2%**	**10.7%**
MSCI World Value Net EUR	*7.0%*	*0.43*	*−54%*	*4.6%*	*9.3%*
MSCI World Net EUR	*8.9%*	*0.58*	*−49%*	*5.7%*	*12.0%*

Net of costs, EUR. 2006-15 and 2016-26: CAGR of each half.

What could go wrong

- A long period of expensive-growth leadership: the strategy will lag MSCI World, as in 2023-24.
- Value traps: eight stocks at 3.75% each, some cheap for good reasons (tobacco, autos).
- Sharp V-shaped recoveries: the risk rule sells low and buys back higher.
- Model risk: rules chosen after several rounds look better on paper than they will live.

The portfolio for July 2026

The last data month is June 2026 and the risk rule is fully invested. The core holds Germany, France, Denmark and Hong Kong, and the Health Care, Consumer Staples, Energy and Real Estate sectors. The satellite holds Enel SpA, British American Tobacco plc, BNP Paribas, Altria Group Inc, Mitsubishi Corp, Rio Tinto plc, The Cigna Group, Mercedes-Benz Group AG. BNP Paribas is both in the satellite and one of the Thoth Ledger stock pitches: the screen found it, the pitch checks whether the discount is a mispricing.

References: Asness, Moskowitz & Pedersen (2013), “Value and Momentum Everywhere”, Journal of Finance 68(3), 929-985 · Faber (2007), “A Quantitative Approach to Tactical Asset Allocation”, Journal of Wealth Management 9(4), 69-79 · Bailey & López de Prado (2014), “The Deflated Sharpe Ratio”, Journal of Portfolio Management 40(5), 94-107.

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